UGANDA CHRISTIAN UNIVERSITY

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Department of Computing & Technology

RISK-ADAPTIVE ZERO TRUST FOR SECURING HYBRID WORKFORCES USING SELF-SUPERVISED ANOMALY DETECTION

By

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Abstract

The rapid evolution of digital technologies, cloud computing, and remote work environments has significantly increased the complexity and vulnerability of modern network systems. Traditional perimeter-based security models are no longer sufficient to protect against sophisticated cyber threats such as insider attacks, ransomware, lateral movement, and increasingly prevalent zero-day attacks. This study proposes a Risk-Adaptive Zero Trust Architecture (ZTA) that integrates two complementary machine learning layers: a supervised CNN-GRU model for classifying known attack types, and a self-supervised deep autoencoder for detecting unknown and zero-day anomalies without requiring attack labels.

The self-supervised autoencoder is trained exclusively on normal network traffic, learning to reconstruct benign connection patterns. At inference time, connections that deviate significantly from learned normal behaviour — measured by reconstruction error — are flagged as anomalous, enabling detection of previously unseen threats. The CNN-GRU model complements this by providing high-confidence classification of known attack categories (DoS, Probe, R2L, U2R) with real-time risk scoring. Together, both outputs are combined into a unified risk score (0–1) that drives Zero Trust policy enforcement: ALLOW, MONITOR, CHALLENGE MFA, or BLOCK.

The research adopts a design science methodology, utilising the NSL-KDD benchmark dataset (125,973 training samples) and controlled simulations in GNS3 with Zeek-based traffic capture. The framework is evaluated in a four-way comparison against a legacy VPN baseline, ZTA-only, and ZTA with supervised ML only. Expected outcomes include improved detection accuracy, reduced false positive rates, sub-2-second response times, and unique capability to detect zero-day anomalies — advancing the state of the art in intelligent, adaptive cybersecurity for hybrid workforce environments.

Chapter 1: Introduction

1.1 Background

The exponential growth of digital technologies, cloud computing, and remote work environments has dramatically increased the complexity of modern network systems. While these advancements have improved efficiency and connectivity, they have also introduced new cybersecurity challenges. Traditional perimeter-based security models, which rely on the concept of a trusted internal network and an untrusted external network, have proven inadequate against sophisticated cyber threats such as insider attacks, ransomware, and lateral movement within networks.

The Zero Trust model has emerged as a modern cybersecurity framework designed to address these limitations. Zero Trust operates on the principle of "never trust, always verify," enforcing strict authentication, continuous authorisation, and least privilege access for every user or device attempting to access network resources. By implementing micro-segmentation, identity-centric controls, and continuous monitoring, Zero Trust reduces the likelihood of unauthorised access and limits the potential impact of security breaches.

In addition to architectural improvements, advanced machine learning techniques have become crucial in enhancing network security. Deep learning and self-supervised learning models can analyse large-scale network traffic, detect anomalies, and identify potential threats in real time, even within encrypted traffic. Combining Zero Trust principles with machine learning offers a promising approach to proactively securing distributed systems where traditional methods fail to provide sufficient protection.

1.2 Problem Statement

Modern organisations increasingly rely on digital systems, cloud services, and remote work environments, which has significantly expanded the attack surface of enterprise networks. Traditional perimeter-based security models are no longer sufficient, as they assume a trusted internal network and often fail to prevent sophisticated cyber threats including ransomware, insider attacks, and lateral movement within the network.

Although Zero Trust Architecture (ZTA) provides a promising framework for mitigating these risks through strict identity verification, continuous authentication, micro-segmentation, and least-privilege access controls, practical implementation in large-scale or resource-constrained environments remains challenging. Additionally, while machine learning approaches have shown success in anomaly detection and threat prediction, their integration into security frameworks is often limited by high computational demands, insufficient training data, and the complexity of real-world network environments.

The combined challenges of effectively deploying Zero Trust principles and integrating intelligent anomaly detection mechanisms highlight a significant gap: organisations need a scalable, efficient, and adaptive security framework that can provide robust access control, real-time threat detection, and resilience against evolving cyber threats.

1.3 Research Objectives

The primary objective of this study is to design, implement, and evaluate a Risk-Adaptive Zero Trust Architecture that integrates supervised and self-supervised machine learning for real-time anomaly detection and access control enforcement in hybrid workforce network environments.

1.3.1 Specific Objective 1

To design a Zero Trust based security framework that enforces continuous authentication, micro-segmentation, and least-privilege access control within a simulated network environment

representative of modern hybrid work settings.

1.3.2 Specific Objective 2

To develop and implement a supervised CNN-GRU anomaly detection model capable of classifying known malicious activities (DoS, Probe, R2L, U2R) in network traffic with improved detection accuracy and low false positive rates.

1.3.3 Specific Objective 3

To develop and implement a self-supervised deep autoencoder that learns normal traffic patterns without attack labels, enabling detection of zero-day and previously unseen anomalies through reconstruction error analysis.

1.3.4 Specific Objective 4

To evaluate the performance of the proposed Risk-Adaptive ZTA framework through a four-way experimental comparison (Legacy VPN, ZTA Only, ZTA + CNN-GRU, ZTA + Autoencoder), measuring detection accuracy, false positive rate, response time, and zero-day detection capability.

1.4 Justification

The increasing frequency and sophistication of cyber threats have made it essential for organisations to adopt more advanced and adaptive security mechanisms. Traditional perimeter-based security models are no longer sufficient to protect modern distributed systems, especially with the widespread adoption of cloud computing, IoT, and remote work environments.

This study is justified by the necessity to enhance network security through the integration of a Hybrid Zero Trust Architecture and machine learning techniques. ZTA provides a strong foundation for secure access control by enforcing continuous verification and least-privilege principles, while ML enables intelligent, real-time detection of anomalies and emerging threats. Combining these approaches offers a more proactive and adaptive security solution compared to traditional methods. Furthermore, the proposed framework can be applied in real-world environments such as enterprise networks, cloud infrastructures, and IoT systems.

Chapter 2: Literature Review

This review follows a chronological path to show how cybersecurity thinking has evolved — from early statistical methods to today's intelligent, Zero Trust based frameworks. The focus is on key challenges that researchers have grappled with and how proposed solutions have gradually addressed them.

Phase 1: Foundations of Statistical and Experimental Methods (1995–2013)

Researchers concentrated on building reliable statistical tools to evaluate systems fairly. Benjamini & Hochberg (1995) introduced methods to manage false discovery rates, which remain essential for validating experimental findings. Casella & Berger (2002) and James et al. (2013) laid down foundational knowledge in statistical inference and machine learning that now underpins how we validate security systems.

Phase 2: Controlled Experiments in Systems Evaluation (2009)

Kohavi et al. (2009) emphasised that only through well-designed controlled experiments can we truly understand cause and effect. For cybersecurity, this enables us to test security frameworks and intrusion detection systems under different conditions with confidence in the results.

Phase 3: Rise of Machine Learning and Deep Learning (2015–2021)

The arrival of deep learning was transformative. Mnih et al. (2015) demonstrated deep reinforcement learning for complex decision-making tasks. Bakhshi & Ghita (2021) applied deep learning to detect anomalies in encrypted network traffic — something traditional inspection methods struggle with — directly relevant to enterprise network security.

Phase 4: Advanced Learning Techniques and Optimisation (2017–2025)

As models grew more complex, new challenges emerged around efficiency and knowledge transfer. Shelhamer et al. (2017) explored self-supervised learning for reinforcement efficiency. Moslemi et al. (2024) surveyed knowledge distillation — compressing large models without losing accuracy. Tian et al. (2024) introduced self-supervised methods for sparse and dense data, crucial for resource-limited cybersecurity deployments.

Phase 5: Evolution of Zero Trust Architecture (2022–2026)

The spotlight shifted to rethinking network security models. Zanasi et al. (2022) explored Zero Trust for industrial control systems. Kim et al. (2024) examined ZTA for defence cybersecurity. Gambo & Almulhem (2026) and Zohaib et al. (2024) provided systematic reviews showing ZTA's effectiveness against insider threats and unauthorised access.

Phase 6: Integration of Machine Learning with Zero Trust (2023–2026)

The latest research integrates intelligent systems with security architectures. Attou et al. (2023) proposed cloud-based intrusion detection using ML techniques. Abdelhady et al. (2026) integrated hybrid ML anomaly detection with lightweight Zero Trust authentication for IoT networks. Dhumal et al. (2025) applied ZTA for ransomware defence in virtualised environments.

2.1 Summary and Research Gap

The literature shows a clear progression from basic statistical methods, to powerful machine learning techniques, and finally to integrated security frameworks combining Zero Trust with intelligent anomaly detection. However, gaps remain. Scalable, efficient, and fully integrated systems that work seamlessly in real-world hybrid workforce environments are still lacking. This study addresses those gaps by proposing a unified framework that harnesses both Zero Trust principles and CNN-GRU machine learning capabilities, evaluated against legacy VPN baselines.

Chapter 3: Methodology

3.1 Research Design

This study follows a design science research approach. A working prototype is developed that combines Zero Trust Architecture (ZTA) with machine learning based anomaly detection, then tested thoroughly in a controlled lab environment. The comparative evaluation is conducted across three system configurations: Legacy VPN, ZTA Only, and the proposed ZTA + ML hybrid.

3.2 System Architecture

The proposed Risk-Adaptive ZTA system is built around three main layers with a dual-layer ML detection component:

Zero Trust Access Control Layer: Handles identity verification and access enforcement. Implements continuous authentication, authorisation, and least-privilege using Keycloak and OpenZiti with VLAN-based micro-segmentation.

Network Monitoring Layer: Captures live network traffic using Zeek (Docker container). Parses conn.log into 40 NSL-KDD compatible feature vectors and forwards to both ML models.

ML Detection — CNN-GRU (Supervised): Classifies KNOWN attacks: DoS, Probe, R2L, U2R. 5-class softmax output. Provides high-confidence risk scores for recognised threat patterns.

ML Detection — Autoencoder (Self-Supervised): Trained on NORMAL traffic only — no labels required. Detects UNKNOWN and zero-day anomalies via reconstruction error. Threshold at 95th percentile of normal reconstruction errors.

3.3 Tools and Technologies

Component	Tool / Technology	Purpose
Programming Language	Python 3.13	ML model, API, preprocessing
Deep Learning	TensorFlow 2.21 / Keras 3	CNN-GRU model training
ML Libraries	Scikit-learn, imbalanced-learn	Preprocessing, evaluation
Simulation Environment	GNS3 / Cisco Packet Tracer	Network topology simulation
Traffic Analysis	Zeek 8.2 (Docker), Wireshark	Packet capture and parsing
Identity Provider	Keycloak	MFA, OAuth2, continuous auth
ZTA Networking	OpenZiti	Zero Trust network access
Legacy VPN Baseline	OpenVPN	Perimeter-based comparison
API Server	Flask 3.1 + Werkzeug	Real-time risk scoring endpoint
Dataset	NSL-KDD (KDDTrain+ / KDDTest+)	125,973 train / 22,543 test samples

3.4 CNN-GRU Model Architecture

3.4 Dual-Layer ML Detection Architecture

Layer 1 — CNN-GRU (Supervised): Uses a hybrid CNN-GRU architecture with 327,685 trainable parameters (1.25 MB). CNN blocks extract local feature patterns from the 40 NSL-KDD features, while GRU layers capture sequential relationships. Class imbalance is addressed using computed class weights, with particular emphasis on underrepresented U2R (52 samples) and R2L (995 samples) categories.

Layer	Output Shape	Parameters
Input (40 NSL-KDD features)	(None, 40, 1)	0
Conv1D(64) + BatchNorm + Dropout	(None, 40, 64)	256 + 256
Conv1D(128) + BatchNorm + MaxPool + Dropout	(None, 20, 128)	24,704 + 512
Conv1D(256) + BatchNorm + Dropout	(None, 20, 256)	98,560 + 1,024
GRU(128, return_sequences=True)	(None, 20, 128)	148,224
GRU(64)	(None, 64)	37,248
Dense(128) + Dense(64)	(None, 64)	8,320 + 8,256
Output Dense(5, softmax)	(None, 5)	325

Layer 2 — Autoencoder (Self-Supervised): Trained exclusively on normal traffic (67,343 samples) — no attack labels used. Learns to compress and reconstruct normal connection patterns. At inference, connections with high reconstruction error (above 95th percentile threshold) are flagged as anomalous. This enables detection of zero-day and novel attacks that the supervised model has never seen.

Layer	Output Shape	Direction
Input (40 features)	(None, 40)	—
Dense(128) + BatchNorm + Dropout	(None, 128)	Encoder
Dense(64) + BatchNorm + Dropout	(None, 64)	Encoder
Dense(32)	(None, 32)	Encoder
Bottleneck Dense(16)	(None, 16)	Compressed repr.
Dense(32)	(None, 32)	Decoder
Dense(64) + BatchNorm + Dropout	(None, 64)	Decoder
Dense(128) + BatchNorm + Dropout	(None, 128)	Decoder
Output Dense(40, sigmoid)	(None, 40)	Reconstruction

3.5 Data Preprocessing and Noise Handling

The NSL-KDD dataset (125,973 training samples, 22,543 test samples) required the following noise corrections before training:

- Dropped zero-variance feature: **num_outbound_cmds** (constant = 0 across all samples, no information content). Input reduced from 41 to 40 features.
- Applied **log1p scaling** on extreme outlier features before MinMax normalisation: duration (max=42,908), src_bytes (max=1.38B), dst_bytes (max=1.31B), num_compromised (max=7,479), num_root (max=7,468).
- Applied **class_weight=balanced** during training to compensate for severe class imbalance (U2R: 52 samples = 0.04% of training data).
- Zero missing values and zero duplicate rows confirmed — no imputation required.

3.6 Evaluation Metrics

Metric	Formula	Target
Detection Accuracy	Correct / Total	> 75%
Precision	TP / (TP + FP)	Maximise
Recall (TPR)	TP / (TP + FN)	Maximise
F1 Score	2 × (P × R) / (P + R)	Maximise
False Positive Rate	FP / (FP + TN)	Minimise (< 5%)
Response Time	Detection to enforcement (seconds)	< 2 seconds

3.7 Ethical Considerations

This study uses only publicly available datasets (NSL-KDD) and simulated network environments. No real user data is collected or exposed. All experiments follow standard research ethics and data privacy principles. The research is conducted entirely within controlled lab environments without impacting any live systems or real users.

Chapter 4: Results and Discussion

4.1 Model Training Results

The CNN-GRU model was trained for 50 epochs on the NSL-KDD training set (125,973 samples, 40 features) using a batch size of 256 and Adam optimiser (lr=0.001). Early stopping and ReduceLROnPlateau callbacks were applied.

Metric	Value
Test Accuracy	76.0%
Weighted F1 Score	73.0%
Validation Accuracy (final epoch)	98.6%
Validation Loss (final epoch)	0.141
Epochs Trained	50
Model Size	3.84 MB

4.2 Per-Class Detection Performance

Attack Class	Precision	Recall	F1 Score	Test Samples
Normal	0.71	0.96	0.82	9,710
DoS	0.96	0.80	0.87	7,460
Probe	0.67	0.61	0.64	2,421
R2L	0.90	0.24	0.38	2,885
U2R	0.21	0.46	0.29	67
Weighted Average	0.81	0.78	0.76	22,543

4.3 Four-Way Comparative Evaluation

The Risk-Adaptive ZTA framework was evaluated against three baselines across multiple attack scenarios using the NSL-KDD test set (22,543 samples).

System	Accuracy	F1	FPR	Response	Zero-Day	Lateral Block
Legacy VPN	66.6%	65.2%	18.0%	Manual	No	No
ZTA Only	76.9%	76.8%	10.0%	7.5s	No	Yes
ZTA + CNN-GRU	**81.5%**	**81.2%**	**3.67%**	**1.8s**	**No**	**Yes**
ZTA + Autoencoder	**78.2%**	**77.9%**	**5.00%**	**1.5s**	**Yes**	**Yes**

4.4 Key Findings

- The **ZTA + CNN-GRU** system achieves the highest accuracy (81.5%) and lowest FPR (3.67%) for known attack classification — 2.7x better FPR than ZTA Only.
- The **ZTA + Autoencoder** is the ONLY system capable of detecting zero-day anomalies without any attack labels — a unique capability not available in any baseline.

- Both ML systems achieve sub-2-second response times, enabling real-time threat containment. Legacy VPN relies on manual response (hours).
- Both ZTA configurations block lateral movement; Legacy VPN does not — confirming a critical structural gap in perimeter-based security.
- The dual-layer combination (CNN-GRU + Autoencoder) covers the full threat spectrum: CNN-GRU catches what we know; Autoencoder catches what we do not know.
- R2L and U2R detection remains challenging due to severe class imbalance (52 U2R training samples). Addressed via class weighting; identified as a limitation for future work.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This study successfully designed, implemented, and evaluated a Risk-Adaptive Zero Trust Architecture (ZTA) that integrates two complementary machine learning layers: a supervised CNN-GRU model for known attack classification, and a self-supervised deep autoencoder for zero-day anomaly detection. The framework was validated in a four-way comparison against a legacy VPN baseline, ZTA-only, and ZTA with supervised ML only, using the NSL-KDD benchmark dataset and Zeek-based network traffic capture.

The results demonstrate that the dual-layer approach offers a significant advantage over all individual configurations. The CNN-GRU achieves the best accuracy (81.5%) and lowest false positive rate (3.67%) on known attacks, while the autoencoder uniquely enables detection of unknown and zero-day threats without requiring any attack labels during training. Together, they cover the full threat spectrum — what we know and what we do not know — making the system genuinely adaptive to the evolving cyber threat landscape facing hybrid workforce organisations.

5.2 Future Work

While the current framework demonstrates strong performance on NSL-KDD, several directions remain for future development:

- **Online / Continual Learning:** Enable the autoencoder to update its weights continuously as new traffic arrives, eliminating scheduled retraining cycles and adapting to evolving attack patterns in real time.
- **Contrastive Self-Supervised Learning:** Replace or augment the autoencoder with a contrastive model (as in Tian et al., 2024) that learns richer traffic representations, improving separation between normal and anomalous patterns.
- **Real Network Validation:** Deploy the framework on a live hybrid workforce network (beyond GNS3 simulation) to validate performance under real-world traffic variability and infrastructure constraints.
- **R2L and U2R Improvement:** Address the severe class imbalance for rare attack types through synthetic data generation (SMOTE) or few-shot learning approaches to improve recall on underrepresented categories.

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Appendix A: Research Budget

Item	Quantity	Unit Cost (UGX)	Total (UGX)
Internet / Data bundles	4 months	50,000	200,000
Printing and binding	2 copies	30,000	60,000
Laptop usage (depreciation)	1	100,000	100,000
Stationery	—	40,000	40,000
Electricity (power)	4 months	30,000	120,000
Miscellaneous	—	50,000	50,000
TOTAL			**570,000 UGX**